

Algebra Sparks

Every graded set, every week — 4 weeks, 20 worksheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Find the Rule

Say the rule out loud in one sentence before you write the next number.

Warm-Up 6 ITEMS

3, 6, 9, 12 — next

5, 10, 15, 20 — next

2, 4, 8, 16 — next

20, 17, 14, 11 — next

1, 4, 7, 10 — next

100, 90, 80 — next

Core GRADED • SHOW YOUR WORKING

1. Rule $\times 3$, starting at 2 — the fourth number (2, 6, 18, 54)

2. 1, 3, 7, 15 — next (Double it and add one)

3. Rule $\times 2 + 1$. Put in 5

4. 4, 9, 14, 19 — the tenth number (Start at 4 and add 5 nine times)

5. 2, 5, 10, 17 — next (The gaps go 3, 5, 7, 9)

Challenge NOT GRADED

C1. Rule $\times 4 - 3$. Put in 7

C2. 6, 11, 16, 21 — the twentieth number ($6 + 5 \times 19$)

Function Machines

In goes a number, out comes a number. The machine never changes its mind halfway.

✦ Warm-Up 6 ITEMS

Machine + 8. In 5

Machine $\times$ 6. In 4

Machine $-$ 7. In 20

Machine $\times$ 10. In 9

Machine $\div$ 2. In 18

Machine + 15. In 15

◆ Core GRADED • SHOW YOUR WORKING

1. Machine $\times$ 3 + 4. In 6 (Multiply first)

2. Machine $\times$ 5 $-$ 2. In 7

3. Machine $\times$ 2 + 9. In 12

4. Machine $\times$ 3 + 4. Out 25 — what went in (Undo the + 4 first)

5. Machine $\times$ 4 $-$ 1. Out 27 — what went in

★ Challenge NOT GRADED

C1. In 2 gives 7, in 3 gives 10, in 4 gives 13. In 10 gives (The rule is $\times$ 3 + 1)

C2. In 1 gives 5, in 2 gives 9, in 3 gives 13. In 12 gives (The rule is $\times$ 4 + 1)

Letters Standing for Numbers

A letter is a number nobody has named yet. Put the number in, then work it out.

Warm-Up 6 ITEMS

$n = 4$. Value of $n + 6$

$n = 4$. Value of $3n$

$n = 7$. Value of $n - 3$

$n = 5$. Value of $2n$

$n = 9$. Value of $n + n$

$n = 6$. Value of $10 - n$

Core GRADED • SHOW YOUR WORKING

1. $n = 5$. Value of $3n + 2$

2. $n = 8$. Value of $2n - 5$

3. $a = 3$, $b = 7$. Value of $4a + b$ (12 and 7)

4. $n = 6$. Value of $n \times n$

5. $n = 12$. Value of $n \div 4 + 5$

Challenge NOT GRADED

C1. $n = 4$. Value of $2n + 3n$ (That's 5n)

C2. $p = 9$. Value of $100 - 8p$

One-Step Equations

Do the same thing to both sides. Type the value of the letter.

Warm-Up 6 ITEMS

$x + 4 = 11$

$x - 5 = 9$

$3x = 21$

$x + 12 = 20$

$6x = 30$

$x \div 2 = 8$

Core GRADED • SHOW YOUR WORKING

1. $x + 17 = 42$ _____

2. $8x = 96$ ($96 \div 8$) _____

3. $x - 23 = 19$ _____

4. $x \div 5 = 13$ _____

5. $4x = 100$ _____

Challenge NOT GRADED

C1. $2x + 5 = 19$ (Take 5 from both sides, then halve)

C2. $3x - 7 = 26$

Guess My Rule & the Math Trail

Capstone day. Every stop on the trail is a problem you wrote yourself.

★ Warm-Up 2 ITEMS

Stops on a ten-stop trail

$$x + 9 = 9$$

◆ Core GRADED • SHOW YOUR WORKING

1. Trail rule $\times 2 + 3$. Stop number 6 gives

.....

2. Ten stops, 3 minutes each — total minutes

.....

★ Challenge NOT GRADED

C1. In 5 gives 23, in 8 gives 35. In 10 gives (The rule is $\times 4 + 3$)

C2. $2x + 2x = 40$ (That's $4x$)

C3. A rule turns 4 into 4 and 6 into 8. Put in 10 ($\times 2 - 4$)

Week 1 Answers

Mission 08 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

1.1 • Find the Rule

OUT OF 11

Warm-Up • 15 | 25 | 32 | 8 | 13 | 70

Core • 54 | 31 | 11 | 49 | 26

Challenge • 25 | 101

1.2 • Function Machines

OUT OF 11

Warm-Up • 13 | 24 | 13 | 90 | 9 | 30

Core • 22 | 33 | 33 | 7 | 7

Challenge • 31 | 49

1.3 • Letters Standing for Numbers

OUT OF 11

Warm-Up • 10 | 12 | 4 | 10 | 18 | 4

Core • 17 | 11 | 19 | 36 | 8

Challenge • 20 | 28

1.4 • One-Step Equations

OUT OF 11

Warm-Up • 7 | 14 | 7 | 8 | 5 | 16

Core • 25 | 12 | 42 | 65 | 25

Challenge • 7 | 11

Fri • Guess My Rule & the Math Trail

OUT OF 4

Warm-Up • 10 | 0

Core • 15 | 30

Challenge • 43 | 10 | 16

What a Letter Is

n is a seat a number sits in. $3n$ is three of them.

Warm-Up 6 ITEMS

$n = 4$. Value of $n + 6$

$n = 4$. Value of $3n$

$n = 7$. Value of $n - 3$

$n = 5$. Value of $2n$

$n = 9$. Value of $n + n$

$n = 6$. Value of $10 - n$

Core GRADED • SHOW YOUR WORKING

1. $n = 5$. Value of $3n + 2$

2. $n = 8$. Value of $2n - 5$

3. $n = 6$. Value of $n \times n$

4. $n = 12$. Value of $n \div 4 + 5$

5. $n = 10$. Value of $4n - 10$

Challenge NOT GRADED

C1. $n = 4$. Value of $2n + 3n$

C2. $p = 9$. Value of $100 - 8p$

Evaluate It

Substitute the number, then work it out. One step at a time.

Warm-Up 6 ITEMS

$n = 2$. Value of $5n$

$n = 3$. Value of $n + 7$

$n = 10$. Value of $n \div 2$

$n = 1$. Value of $9n$

$n = 0$. Value of $6n$

$n = 8$. Value of $n - 8$

Core GRADED • SHOW YOUR WORKING

1. $n = 7$. Value of $4n + 1$

2. $n = 6$. Value of $5n - 12$

3. $n = 9$. Value of $2n + 2$

4. $n = 11$. Value of $3n - 3$

5. $n = 4$. Value of $25 - 3n$

Challenge NOT GRADED

C1. $n = 12$. Value of $2n \div 3$

C2. $n = 5$. Value of $n^2 + n$

Write the Expression

Turn the sentence into symbols. Then check it on a number.

✦ Warm-Up 6 ITEMS

Three lots of n , when $n = 4$

n more than 5, when $n = 3$

Double n , when $n = 7$

n less than 10, when $n = 4$

Half of n , when $n = 8$

n plus n , when $n = 5$

◆ Core GRADED • SHOW YOUR WORKING

1. Five more than double n , when $n = 6$

2. Three less than four lots of n , when $n = 5$

3. Ten more than half of n , when $n = 20$

4. Triple n , plus 4, when $n = 8$

5. n multiplied by itself, when $n = 7$

★ Challenge NOT GRADED

C1. Twice the sum of n and 3, when $n = 5$

C2. The perimeter of a square of side s , when $s = 7$

Two Letters

Substitute both before you simplify anything.

Warm-Up 6 ITEMS

$a = 2, b = 3$. Value of $a + b$

$a = 2, b = 3$. Value of ab

$a = 5, b = 1$. Value of $a - b$

$a = 4, b = 4$. Value of $a + b$

$a = 3, b = 6$. Value of $b \div a$

$a = 1, b = 9$. Value of ab

Core GRADED • SHOW YOUR WORKING

1. $a = 3, b = 7$. Value of $4a + b$

2. $a = 4, b = 6$. Value of $3a + 2b$

3. $a = 5, b = 2$. Value of $a \times b + 3$

4. $a = 6, b = 3$. Value of $2a - b$

5. $a = 8, b = 2$. Value of $a \div b + 5$

Challenge NOT GRADED

C1. $a = 4, b = 5$. Value of $2a + 3b - 10$

C2. $l = 7, w = 3$. Value of $2l + 2w$

Perimeter as a Rule

Write it as 4s once, then test it on three squares.

★ Warm-Up 2 ITEMS

Perimeter of a square of side 3

Of side 5

◆ Core GRADED • SHOW YOUR WORKING

1. Using 4s with $s = 9$ — the perimeter

.....

2. A square of perimeter 28 — its side

.....

★ Challenge NOT GRADED

C1. Using $2l + 2w$ with $l = 8$, $w = 5$

.....
C2. A rectangle of perimeter 30 and length 10 — its width

.....
C3. Using 4s, the side when the perimeter is 100

.....

Week 2 Answers

Mission 08 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

2.1 • What a Letter Is

OUT OF 11

Warm-Up • 10 | 12 | 4 | 10 | 18 | 4

Core • 17 | 11 | 36 | 8 | 30

Challenge • 20 | 28

2.2 • Evaluate It

OUT OF 11

Warm-Up • 10 | 10 | 5 | 9 | 0 | 0

Core • 29 | 18 | 20 | 30 | 13

Challenge • 8 | 30

2.3 • Write the Expression

OUT OF 11

Warm-Up • 12 | 8 | 14 | 6 | 4 | 10

Core • 17 | 17 | 20 | 28 | 49

Challenge • 16 | 28

2.4 • Two Letters

OUT OF 11

Warm-Up • 5 | 6 | 4 | 8 | 2 | 9

Core • 19 | 24 | 13 | 9 | 9

Challenge • 13 | 20

Fri • Perimeter as a Rule

OUT OF 4

Warm-Up • 12 | 20

Core • 36 | 7

Challenge • 26 | 5 | 25

Keep It Level

Take the same amount from both sides and the balance holds.

✦ Warm-Up 6 ITEMS

$x + 4 = 11$

$x + 5 = 12$

$x + 10 = 15$

$x + 2 = 9$

$x + 7 = 7$

$x + 12 = 20$

◆ Core GRADED • SHOW YOUR WORKING

1. $x + 17 = 42$

2. $x - 5 = 9$

3. $x - 23 = 19$

4. $x + 38 = 100$

5. $x - 14 = 27$

★ Challenge NOT GRADED

C1. $x + 156 = 400$

.....

C2. $x - 99 = 101$

.....

Multiply and Divide

Divide both sides by the same number and the balance still holds.

✦ Warm-Up 6 ITEMS

$3x = 21$

$6x = 30$

$2x = 14$

$5x = 25$

$x \div 2 = 8$

$4x = 20$

◆ Core GRADED • SHOW YOUR WORKING

1. $8x = 96$

2. $7x = 84$

3. $x \div 5 = 13$

4. $4x = 100$

5. $9x = 108$

★ Challenge NOT GRADED

C1. $12x = 288$

.....

C2. $x \div 7 = 21$

.....

Check Your Answer

Put it back in. If both sides match, you are done — no marking needed.

Warm-Up 6 ITEMS

$x = 7$ in $x + 4$ — the value

$x = 5$ in $3x$ — the value

$x = 9$ in $x - 2$

$x = 4$ in $2x + 1$

$x = 6$ in $x \div 2$

$x = 10$ in $x + 10$

Core GRADED • SHOW YOUR WORKING

1. Solve then check: $3x = 27$

.....

2. Check $x = 9$ in $3x$ — the value

.....

3. Solve then check: $x + 19 = 31$

.....

4. Check $x = 12$ in $x + 19$

.....

5. Solve: $6x = 72$

.....

Challenge NOT GRADED

C1. $2x + 5 = 19$

.....

C2. Check $x = 7$ in $2x + 5$

.....

Word to Equation

Turn the situation into an equation, then solve it.

Warm-Up 6 ITEMS

A number plus 5 is 12. The number

Double a number is 14. The number

A number less 3 is 8. The number

Three lots of a number is 18. The number

Half a number is 9. The number

A number plus itself is 16. The number

Core GRADED • SHOW YOUR WORKING

1. Four friends share equally and get 13 each. The total

.....

2. After spending \$18 you have \$27. You started with

.....

3. Five boxes hold 60 altogether. Each box

.....

4. A number times 7 is 91. The number

.....

5. Twice a number plus 3 is 21. The number

.....

Challenge NOT GRADED

C1. Three times a number, less 7, is 26. The number

.....

C2. A rectangle of perimeter 34 and width 5 — its length

.....

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

✦ Warm-Up 2 ITEMS

6, 11, 16, 21 — next

$n = 4$. Value of $3n$

◆ Core GRADED • SHOW YOUR WORKING

1. Machine $\times 4 + 2$. In 9

.....

2. That machine gives 30 — what went in

.....

3. $n = 7$. Value of $3n - 4$

.....

4. $x + 18 = 45$

.....

5. $7x = 63$

.....

★ Challenge NOT GRADED

C1. A pattern starts at 3 and adds 4 — its tenth number

.....

Week 3 Answers

Mission 08 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

3.1 • Keep It Level

OUT OF 11

Warm-Up • 7 | 7 | 5 | 7 | 0 | 8

Core • 25 | 14 | 42 | 62 | 41

Challenge • 244 | 200

3.2 • Multiply and Divide

OUT OF 11

Warm-Up • 7 | 5 | 7 | 5 | 16 | 5

Core • 12 | 12 | 65 | 25 | 12

Challenge • 24 | 147

3.3 • Check Your Answer

OUT OF 11

Warm-Up • 11 | 15 | 7 | 9 | 3 | 20

Core • 9 | 27 | 12 | 31 | 12

Challenge • 7 | 19

3.4 • Word to Equation

OUT OF 11

Warm-Up • 7 | 7 | 11 | 6 | 18 | 8

Core • 52 | 45 | 12 | 13 | 9

Challenge • 11 | 12

Fri • Mid-Unit Quiz

OUT OF 7

Warm-Up • 26 | 12

Core • 38 | 7 | 17 | 27 | 9

Challenge • 39

Design the Trail

Ten stops, one per mission plus two of your own. Map it first.

✦ Warm-Up 6 ITEMS

Stops on the trail

Missions in the year

Stops of your own choosing

3 minutes per stop, 10 stops — minutes

Half of 10 stops

Two stops per room — rooms needed

◆ Core GRADED • SHOW YOUR WORKING

1. Trail rule $\times 2 + 3$ — stop 6 gives

.....

2. That rule — stop 10 gives

.....

3. 10 stops at 4 minutes each — minutes

.....

4. If 3 stops are outdoors, the percent indoors

.....

5. Stops numbered 1 to 10 — their total

.....

★ Challenge NOT GRADED

C1. A trail rule $\times 3 + 1$ — stop 12 gives

.....

C2. Ten stops, each worth 12 points — the total

.....

Write the Problems

Every stop gets a problem and a hidden answer key in your handwriting.

✦ Warm-Up 6 ITEMS

23×14

$372 \div 3$

$2 + 3 \times 4$

$\frac{1}{2} + \frac{1}{4}$ — type as a/b

$\$20 - \13.68

Perimeter of a 4 by 3 rectangle

◆ Core GRADED • SHOW YOUR WORKING

1. Mean of 4, 7, 7, 9, 13
.....

2. $x + 17 = 42$
.....

3. Area 48, one side 6 — the other
.....4. GCF of 24 and 36
.....5. Round 4,829 to the nearest thousand
.....

★ Challenge NOT GRADED

C1. $\frac{2}{3} + \frac{3}{4}$ — type as a/b
.....

C2. A machine $\times 4 + 3$ — in 10 gives
.....

Mom Walks It

You mark her work. Any ambiguous question gets rewritten on the spot.

Warm-Up 6 ITEMS

10 stops, 8 correct — the percent

10 stops, 9 correct — the percent

10 stops, 10 correct — the percent

10 minus 7

Half of 10 stops

10 stops, 5 correct — the percent

Core GRADED • SHOW YOUR WORKING

1. She takes 4 minutes a stop — total minutes

2. Two stops rewritten out of 10 — the percent

3. 85% of 10 stops — how many

4. If 9 of 10 are solvable, the percent that are not

5. A 40-minute walk over 10 stops — mean minutes per stop

Challenge NOT GRADED

C1. 12 points per stop, 10 stops, she scores 96 — the percent

C2. To score 85% of 120 points, she needs

Year-End Sweep

All eight error journals. Name the one habit that fixed itself.

✦ Warm-Up 6 ITEMS

7×8

$84 \div 12$

$2 + 3 \times 4$

$\frac{1}{2}$ of 20

$0.25 + 0.25$

Degrees in a right angle

◆ Core GRADED • SHOW YOUR WORKING

1. 38×24 _____

2. $618 \div 6$ _____

3. $\frac{2}{3} + \frac{1}{4}$ — type as a/b _____

4. Mean of 6, 8, 10, 12 _____

5. $3x = 21$ _____

★ Challenge NOT GRADED

C1. Primes below 100 — how many

C2. $2x + 5 = 19$

Mission 08 Test

Twelve items plus the completed trail. Final trophy band awarded.

✦ Warm-Up 2 ITEMS

3, 6, 9, 12 — next

$n = 5$. Value of $2n$

◆ Core GRADED • SHOW YOUR WORKING

1. 2, 6, 18, 54 — next

2. A pattern from 5 adding 6 — its twelfth number

3. Machine $\times 5 + 3$. In 8

4. Machine $\times 6 - 4$ gives 32 — what went in

5. $n = 9$. Value of $4n + 7$

6. $a = 4$, $b = 6$. Value of $3a + 2b$

7. $x - 26 = 19$

8. $9x = 108$

★ Challenge NOT GRADED

C1. $2x + 5 = 19$

C2. A machine turns 5 into 23 and 8 into 35 — in 10 gives

Week 4 Answers

Mission 08 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

4.1 • Design the Trail

OUT OF 11

Warm-Up • 10 | 8 | 2 | 30 | 5 | 5

Core • 15 | 23 | 40 | 70 | 55

Challenge • 37 | 120

4.2 • Write the Problems

OUT OF 11

Warm-Up • 322 | 124 | 14 | $\frac{3}{4}$ | 6.32 | 14

Core • 8 | 25 | 8 | 12 | 5000

Challenge • $\frac{17}{12}$ | 43

4.3 • Mom Walks It

OUT OF 11

Warm-Up • 80 | 90 | 100 | 3 | 5 | 50

Core • 40 | 20 | 8.5 | 10 | 4

Challenge • 80 | 102

Thu • Year-End Sweep

OUT OF 11

Warm-Up • 56 | 7 | 14 | 10 | 0.5 | 90

Core • 912 | 103 | $\frac{11}{12}$ | 9 | 7

Challenge • 25 | 7

Fri • Mission 08 Test

OUT OF 10

Warm-Up • 15 | 10

Core • 162 | 71 | 43 | 6 | 43 | 24 | 45 | 12

Challenge • 7 | 43